MSAS – Assignment #2: Modeling

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1 Exercise 1

To perform a preliminary analysis and evaluate a proposed satellite configuration, it is requested to model an active thermal control system and its associated mechanisms to assess the system's capability to maintain the temperatures within a certain range $[T_{min}; T_{max}]$. The satellite body is a rectangular cuboid (1.5 [m] height, 0.5 [m] side), with fixed solar panels (0.5 [m] side, 0.95 [m] length) located on the +Y-axis. The satellite flies with a fixed inertial attitude, with the satellite's Y-axis always directed towards the Sun. The satellite has two extendable radiators, which are located on the ±X-axis faces. The radiators are connected to the satellite body with electrically controlled hinges that can rotate along the Z-axis direction. The satellite with its radiators in open and closed configurations is represented in Figure 1.

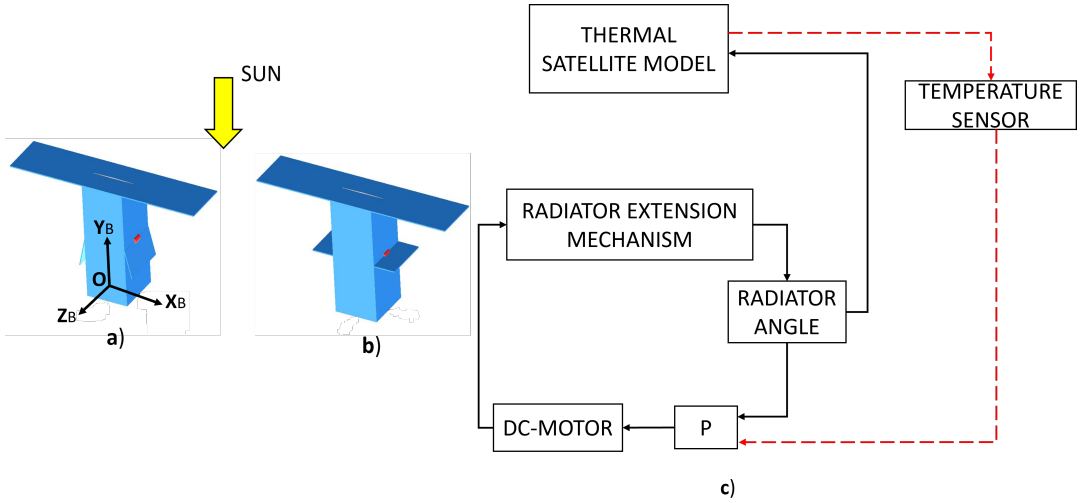


Figure 1: Satellite configuration: **a**) Closed radiator; **b**) Open radiator; **c**) Thermal control logic overview

The extension mechanism is based on a DC motor which gives the needed torque to move the radiator to the desired position. In Figure 2 the physical model of the mechanism is shown. The angle θ represents the rotation of the radiator with respect to the Z-axis and is measured counter-clockwise. For a rotation angle $\theta = 0$ deg, the radiators are in fully open configuration, for $\theta = -0.4\pi$ the radiators are in stowed (closed) configuration. The main parameters of the electro-mechanical model are reported in Table 1.

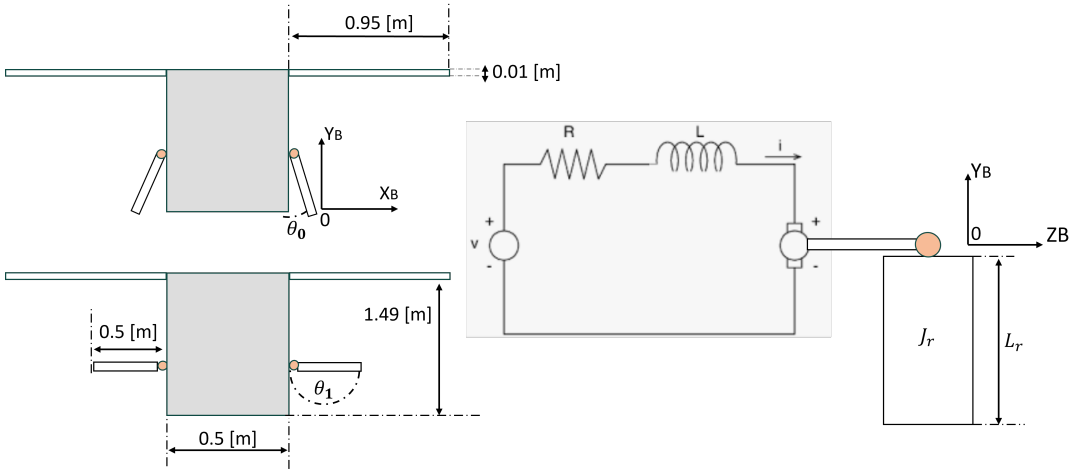


Figure 2: Schematics of the radiator extension mechanism

Parameter	Symbol	Value	Units
Resistance	R	0.1	Ω
Inductance	L	0.001	H
Motor constant	k_m	0.3	Nm/A
Radiator mass	m_r	0.2	kg
Radiator length	L_r	0.5	m

Table 1: Characteristics of the thermal control mechanism.

The radiator has a lower emissivity side, facing deep space in the closed configuration, and a higher emissivity side which is hidden when the radiator is in the stowed position. It is assumed that radiative heat exchange among the satellite surfaces (re-radiation) is negligible and that no other thermal radiation sources (e.g., planets) are present. As a result, the thermal control of the satellite is regulated mainly by the radiator angle θ, as shown in Figure 1 c). By opening and closing the radiator, the effective area of the high- and low-emissivity sides exposed to deep space increases and decreases respectively. This results in a linear variation of the emissivity ϵ as a function of θ that can be expressed as in Eq. (1):

$$\epsilon(\theta) = \epsilon_{min} + \left[\frac{\epsilon_{max} - \epsilon_{min}}{0.4 \cdot \pi}\right](\theta(t) + 0.4 \cdot \pi) \tag{1}$$

When the radiator is in a closed configuration with $\theta(t) = -0.4\pi$ [rad] the emissivity is at its minimum $\epsilon_{min} = 0.01$. In fully open configuration instead, when $\theta(t) = 0$ [rad], the radiator emissivity reaches its maximum value $\epsilon_{max} = 0.98$. Using a lumped nodes thermal modeling, the satellite can be divided into 5 components (the two solar panels, the main body, and the two radiators). The radiators and solar panels are connected to the main body node by a conductive path, while all nodes are thermally radiating towards deep space (see Figure 3).

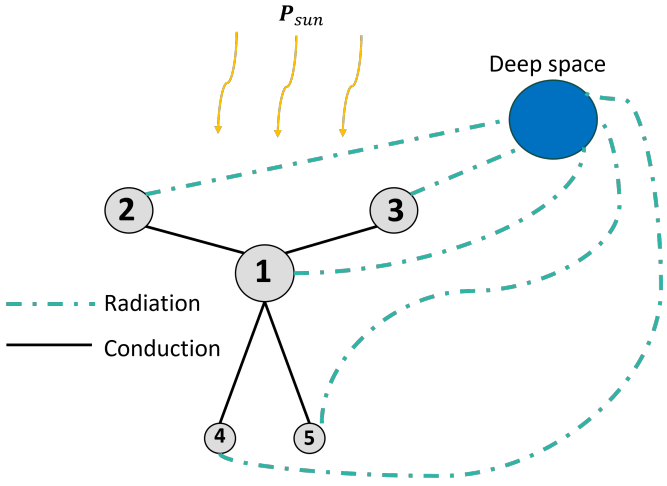


Figure 3: The thermal lumped nodes model

The solar radiation from the Sun is acting only on the two solar panel nodes and on the top face of the main-body node (the radiators are always in shadow). The temperature of deep space is constant and equal to T_{ds}= 3 [K]. The characteristics of the thermal network and its components are given in Table 2.

Parameter	Symbol	Value	Unit
Sun Power	P_{Sun}	1350	W/m^2
Heat capacity	C_1	1.5×10^5	J/K
Heat capacity	C_2, C_3	1187.5	J/K
Heat capacity	C_4, C_5	30	J/K
Thermal conductance	G_{12}, G_{13}, G_{14}, G_{15}	10	W/K
Absorptivity	α_1	0.6	-
Absorptivity	α_2, α_3	0.78	-
Emissivity	ϵ_1	0.45	-
Emissivity	ϵ_2, ϵ_3	0.75	-

Table 2: Thermal parameters of the system.

Part 1: causal modeling (9 points)

Considering the following main constitutive equations of the system:

$$C_i \frac{dT_i}{dt} = \sum_i^n (Q_{in}^i - Q_{out}^i) \tag{2}$$

$$V_{in} = k_p \cdot (T_1 - T_1^{ref}) \tag{3}$$

$$J_r \ddot{\theta} = \tau_{in} \tag{4}$$

where Eq. (2) is the conservation of energy at a thermal node; Eq. (3) is a proportional control law in which V_{in} is the input voltage applied on the DC-motor, and T_1^{ref} is the reference temperature of the main-body; and Eq. (4) is the simplified rotational radiator dynamics where the input torque is linked to current flowing into the motor through the relation: $\tau_{in} = k_m \cdot i$. Assuming that: the upper limit $\hat{T}_{max}$ and lower limit temperature $\hat{T}_{min}$ allowed for **node 1** are 300 [K] and 290 [K] respectively, the motor control is capable of maintaining the angle θ within the physical limits, all the thermal nodes start with a temperature of 298.15 [K], and that the radiator is closed at the beginning of the simulation ($\theta(0) = -0.4\pi$):

1. Formulate the full system of non-linear ODEs making explicit the state variables of the whole simulation.
2. Considering a simulation time of at least 50 [h], define the value for the proportional gain k_p such that: the temperature is kept around the target $T_1^{ref} = 294.15$ [K], and within $t = 10$ [h] the maximum temperature oscillations are less than 0.1% of T_1^{ref}. Plot the resulting temperature evolution over time on all thermal nodes.
3. Discuss the results and the ode used in Matlab for the integration.

Part 2: acausal modeling (6 points)

Using the Modelica standard library, reproduce in Dymola the physical model of the thermal-electro-mechanical system described above. You can build your **own model block** in Modelica to implement specific signals such as $\epsilon(\theta)$ or the control strategy. Please note that when you create a new model block (see in Table 3) you can insert input/output variables and your equations (see as an example the **Text View** of the block: *Modelica/Blocks/Math/Add*). Then, simulate it in OpenModelica comparing the results with the ones obtained using the causal model.

(15 points)

Physical element	*Modelica library*
Rigid-body	Modelica/Mechanics/MultiBody/Parts
Revolute joint	Modelica/Mechanics/MultiBody/Parts
Fixed-translational element	Modelica/Mechanics/MultiBody/Parts
DC-motor	Modelica/Electrical/Machines/BasicMachines/DCMachines
Thermal node	Modelica/Thermal/HeatTransfer/Components
Thermal resistances	Modelica/Thermal/HeatTransfer/Components
Logical switch	Modelica/Blocks/Logical/Switch
Hysteresis	Modelica/Blocks/Logical/Switch
Temperature sensor	Modelica/Thermal/HeatTransfer/Sensors
Your model block	File/New/Block

Table 3: Modelica libraries.

1.1 Casual modeling

1.1.1 ODEs formulation

The system of non-linear ODEs has been formulated using 8 state variables: the temperature of the S/C body (T_1), the temperatures of the two solar panels (T_2, T_3), the temperatures of the two radiators (T_4, T_5), the current flowing into the electric circuit (i), the radiator angle (θ) and the angular velocity of the radiator ($\dot{\theta}$).
To write the non-linear ODEs of the thermal network, first is necessary to identify the thermal contributions defined in Equation 5, and divide them between input and output.

$$Q^i_{in} = Q_{Sun,i} = P_{Sun} A_i \alpha_i \qquad Q^i_{out} = \begin{cases} Q_{cond,i \leftrightarrow j} = G_(T_i - T_j) \\ Q_{rad} = \sigma A_i \epsilon_i (T_i^4 - T_{DS}^4) \end{cases} \tag{5}$$

Then is possible to write the energy balance for all five thermal nodes, it is necessary to point out that the temperature evolution and so the energy balance for the two solar panels is equal, the same for the two radiators.

$$C_i \frac{dT_i}{dt} = \sum_i^n (Q^i_{in} - Q^i_{out}) \tag{6}$$

The circuit equation has been obtained by applying Kirchoff's law to the circuit

$$V_{in} = v_R + v_L + v_m \tag{7}$$

where v_R, v_L, v_m are the voltage of the resistor, the inductor and the motor, respectively.

$$v_R = Ri \quad v_L = L\frac{di}{dt} \quad v_m = k_m \dot{\theta} \tag{8}$$

After assembling all the ODEs and making explicit the state variables, the system reported in Equation 9 is obtained.

$$\begin{cases} \dot{T_1} = \dfrac{P_{Sun} A_{1,a} \alpha_1}{C_1} - \dfrac{\sigma A_{1,e} \epsilon_1 (T_1^4 - T_{DS}^4)}{C_1} - \dfrac{G}{C_1}(4T_1 - T_2 - T_3 - T_4 - T_5) & \\ \dot{T}_{SolarPanel,j} = \dfrac{P_{Sun} A_{j,a} \alpha_j}{C_j} - \dfrac{\sigma A_{j,e} \epsilon_j (T_j^4 - T_{DS}^4)}{C_j} - \dfrac{G}{C_j}(T_j - T_1) & \text{for } j = 2,3 \\ \dot{T}_{Radiator,l} = -\dfrac{\sigma A_{l,e} \epsilon(\theta) (T_l^4 - T_{DS}^4)}{C_l} - \dfrac{G}{C_l}(T_l - T_1) & \text{for } l = 4,5 \\ \\ \dot{i} = \dfrac{k_p (T_1 - T_{1,ref})}{L} - \dfrac{Ri}{L} - \dfrac{k_m \dot{\theta}}{L} & \\ \dot{\theta} = \dot{\omega} & \\ \dot{\omega} = \dfrac{k_m i}{J_r} & \end{cases} \tag{9}$$

Where $A_{i,a}$ and $A_{i,e}$ are the absorption and emission area of the i-th thermal node;

1.1.2 k_p computation and θ limitation

In order to achieve the performances requested, $T_1 \in [T_{min}, T_{max}]$ and reach the $0.1\% T_{1,ref}$ in $10\,h$, k_p must tuned accurately, in this case it has been set equal to $1.2e-4$.
By tuning k_p is not possible to reach the performances without having θ that execs bounds, so it has been necessary the implement of a control logic. Different solutions have been checked. The main two are:

- `if` $(\theta > 0$ `&&` $\dot{\theta} > 0$ `||` $\theta < -0.4\pi$ `&&` $\dot{\theta} < 0) \implies \dot{\theta} = 0$
- `if` $(\theta > 0$ `&&` $T_1 > T_{1,ref}$ `||` $\theta < -0.4\pi$ `&&` $T_1 < T_{1,ref}) \implies V_{in} = 0$

Both bounds the angle of the radiator and allow the system to reach the performances requested, the difference is in the solutions of i and $\dot{\theta}$, in fact, for the first control if θ remains constant both values do not stay at zero, and this is not a correct representation of the physical system, considering that the second control logic has been chosen.

1.1.3 Results

The system has been solved using a stiff ODE solver,`ode23s` it uses more steps than `ode15s` but no informations are lost.
By analyzing Figure 4 is possible to see the behaviors of the temperature profiles for each component, as specified before due to the symmetric nature of the problem the temperature profile of the right solar panel is the same as the left one, the same happens for the radiators. As expected they start oscillating until they reach their asymptotic values.
Figure 5 shows T_1, from which it is possible to observe that the constraints have been respected.

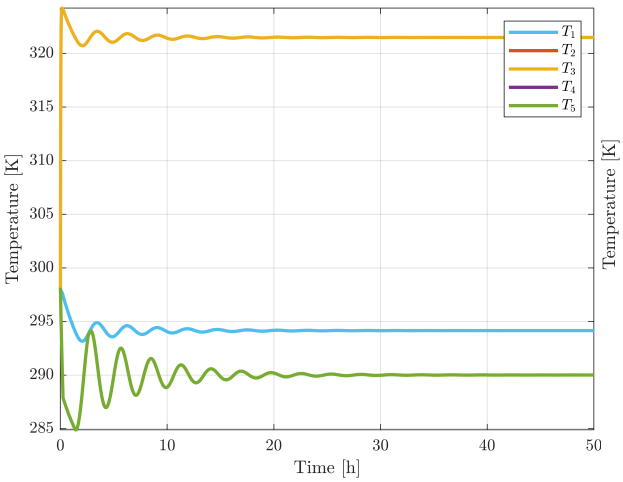


Figure 4: Temperatures evolution

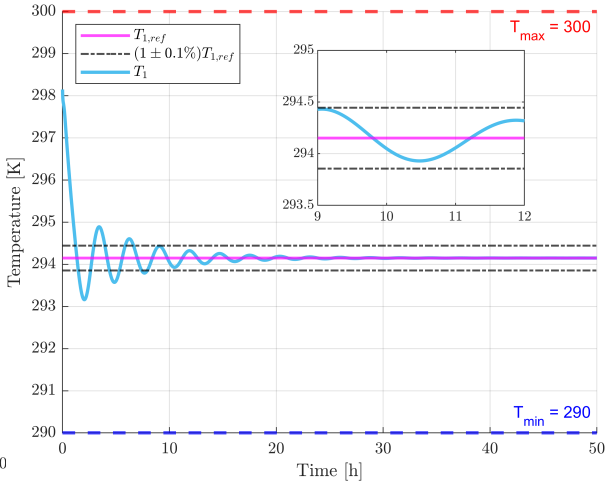


Figure 5: Evolution of the temperature of the S/C body

By looking at the evolution in time of the other state variables Figure 6, it is possible to notice that θ stays within the boundary, as for the temperature profiles it starts oscillating until at the end of the simulations it stabilizes around the equilibrium value. Can be observed that the and the radiators stay fully open approximately between $0.25\,h$ and $1.35\,h$, while staying fully close between $2.6\,h$ and $2.8\,h$, roughly.

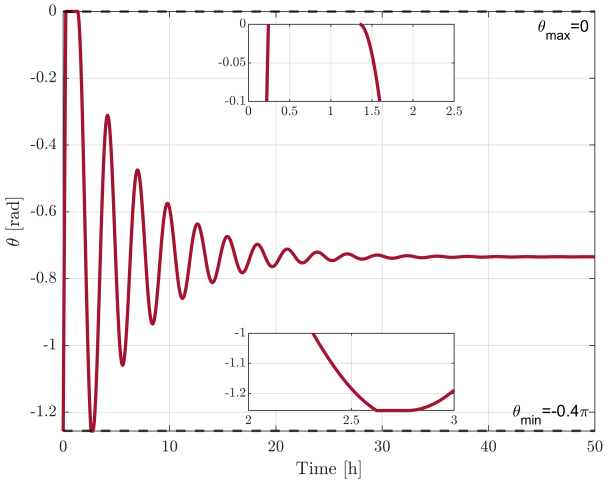


Figure 6: Radiator angle evolution

$\dot{\theta}$ and i in Figure 7 and Figure 8 respectively show a coherent behavior with the one of θ. In fact, after the starting oscillating behavior, can be observed that in the period of time in which the angle is constant both the angular velocity and the current go to zero. At the end of the simulation, they tend asymptotically to zero due to the fact that theta tends to a constant value.

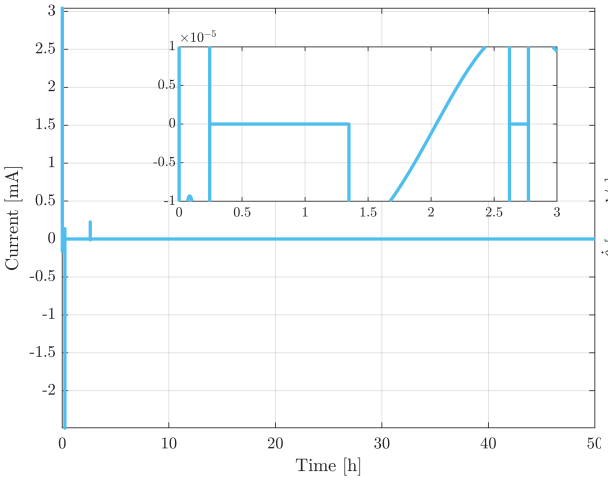


Figure 7: Current evolution

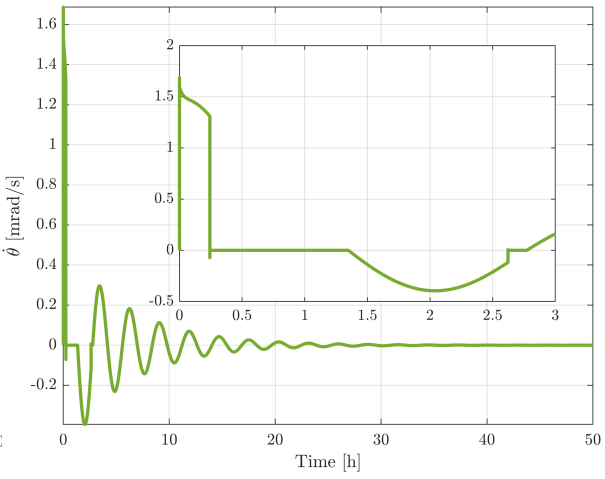


Figure 8: Radiator angular velocity evolution

1.2 Acausal modeling

The physical model developed in *Dymola* is represented in Figure 9. The subscripts (_L) and (_R) in the block representation, stand for Left and Right respectively.
For the thermal network: the Sun heat flux has been modeled using *FixedHeatFlow*, the conduction and radiation have been modeled using the *ThermalConductor* and *BodyRadiation* block respectively, the only difference stand for the radiation between the radiators and the deep space, in fact, a radiation block with which takes in input also emissivity is not present, so it has been models using *Convection* block, which takes in input h_{Rad}, computed using Equation 10, to avoid complexity in graphic model the equation of h_{rad} has been developed inside the h_Rad_R block.

$$h_{rad,l} = \sigma A_{l,e} \epsilon(\theta)(T_l^2 + T_{DS}^2)(T_l + T_{DS}) \qquad \text{for } l = 4,\ 5 \tag{10}$$

The control law has been modeled using *Gain* and *Sum* block, then V_{in} computed goes in input in the block for the limitation of θ and then using the condition discussed above the voltage goes as input in the circuit model.
For what concerning the circuit has been modeled using *SignalVoltage*, *Resistor*, *Inductor*, and *RotationalEMF*. The torque that comes out from the rotational motor, goes as input in the radiator's rotational inertia, to obtain θ the *AngleSensor* is used.
The radiator angle then becomes the input for the *Emissivity* block necessary as input in the *h_Rad_R* block. As for h_{rad} also the emissivity equation and the control logic to limit θ have been developed inside the other two blocks, *Emissivity* and *ControlBlock* respectively. The model has been simulated in *OpenModelica*, and the results obtained are very similar to

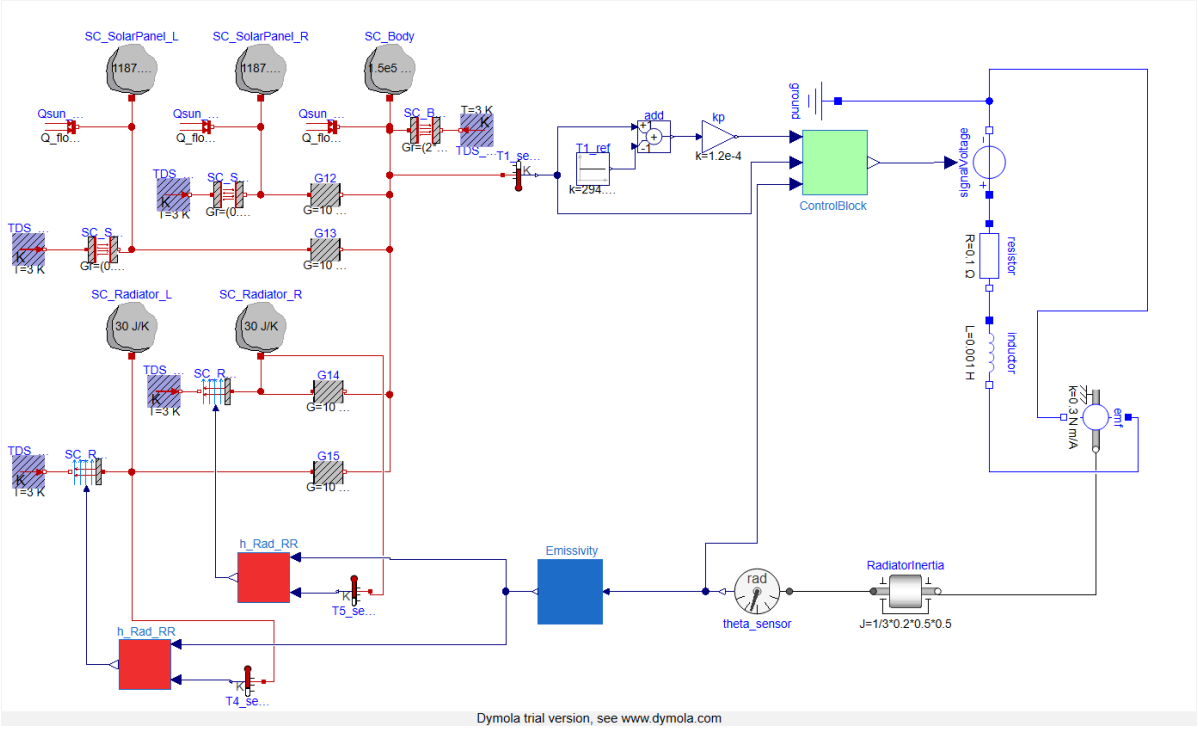


Figure 9: Dymola model

the MATLAB one discussed above. There are small differences related to the difference in the integrator, but the overall behavior is the same. Figure 10 shows the evolution of all the temperatures, and Figure 11 shows the temperature of the S/C body, also, in this case, it respects the constraints.

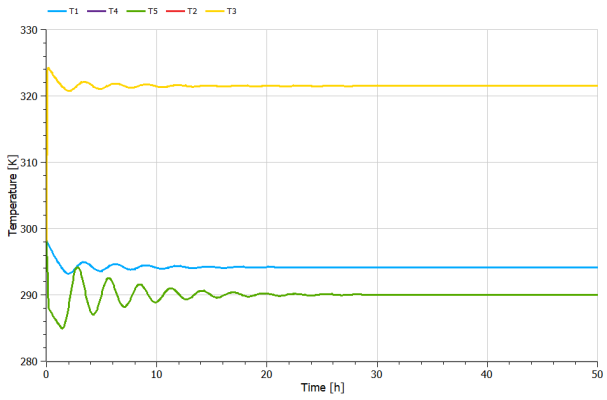
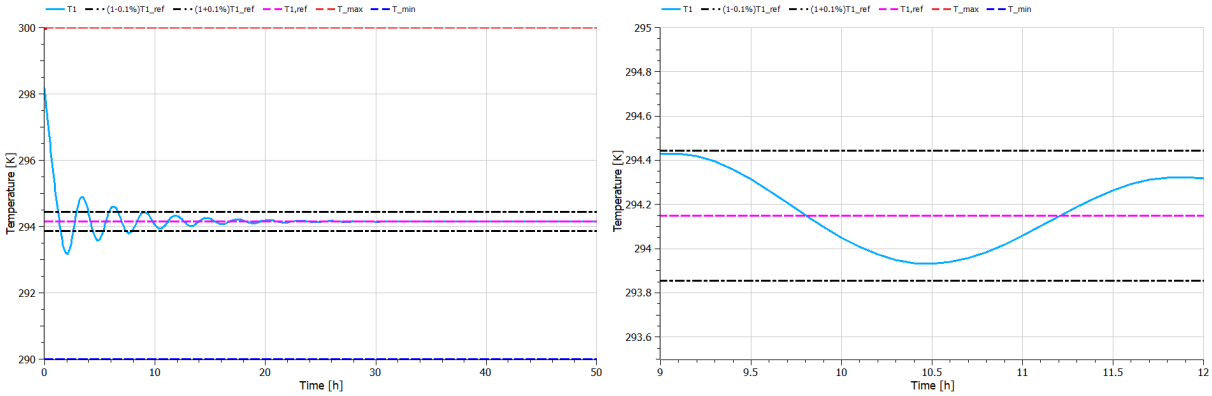


Figure 10: Temperatures evolution



(a) T_1 **(b)** Detailed T_1

Figure 11: Evolution of the temperature of the S/C body

θ and $\dot{\theta}$ shown inFigure 12 and Figure 13 have the same behavior of the MATLAB solution. OpenModelica is able to represent correctly the evolution $\dot{\theta}$ when θ is constant.

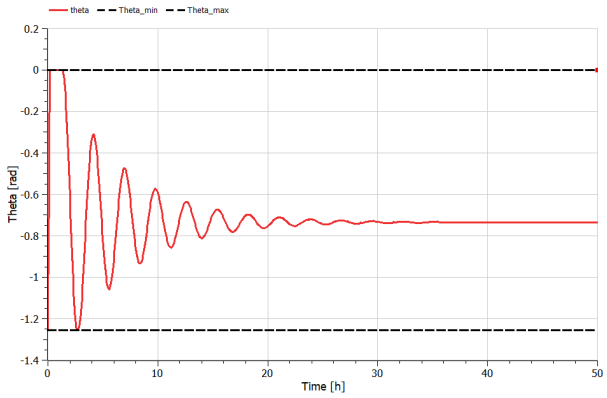
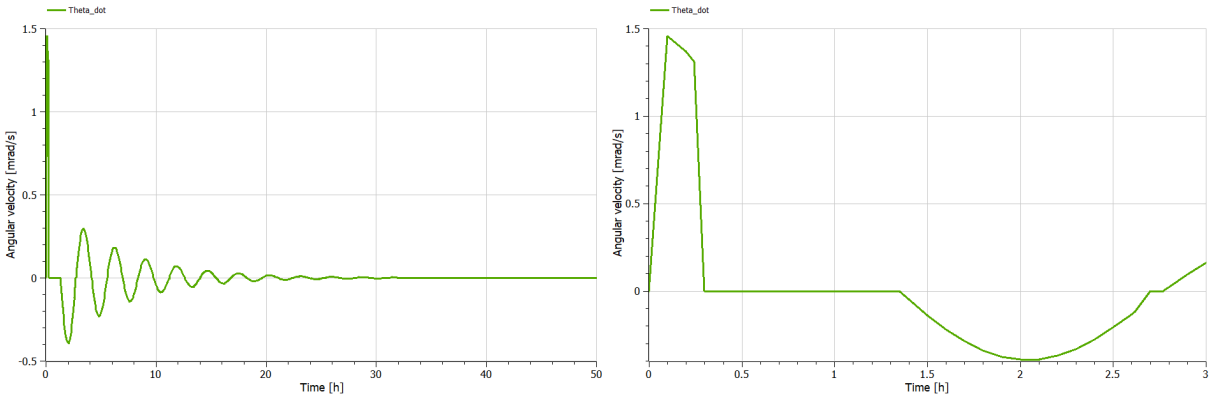


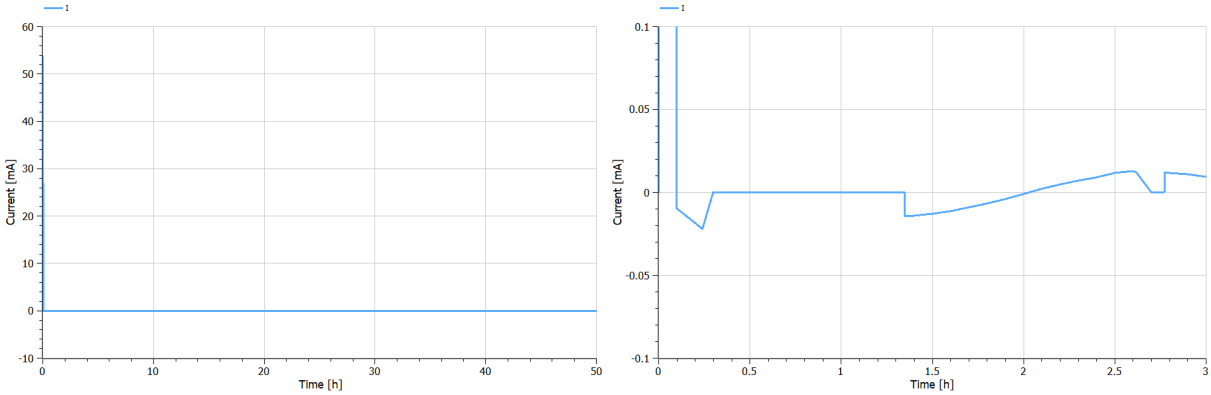
Figure 12: Radiator angle evolution



(a) $\dot{\theta}$ **(b)** Detailed $\dot{\theta}$

Figure 13: Radiator angular velocity evolution

For I the main difference is the absence of the negative oscillations at the beginning of the simulation, and also the amplitude of the peaks at the beginning and when the current stops due to the saturation of θ are different.



(a) I **(b)** Detailed I

Figure 14: Current evolution

As discussed above and by looking at the detailed picture for $\dot{\theta}$ and I, Figure 13 and Figure 14 respectively, is possible to notice that the solutions computed with OpenModelica are not as accurate as the ones obtained with MATLAB, this loss in accuracy does not allow the tracking of fast transient behavior, as happened for both $\dot{\theta}$ and I.

2 Exercise 2

A simplified conceptual map of the Attitude Control System (ACS) of a satellite is depicted in Figure 15. For the accuracy of some measurements that the spacecraft has to obtain, the probe has to fly in an orbit very close to the Earth's surface. The spacecraft is therefore subjected to a high atmospheric drag. The ACS has to compensate for the deceleration caused by this drag continuously. In general, ACSs are made of 3 sensors in orthogonal directions and 3 thrusters to detect and compensate for linear and angular accelerations on the probe in each direction, but for the sake of this project, only compensation in the tangential direction will be considered.

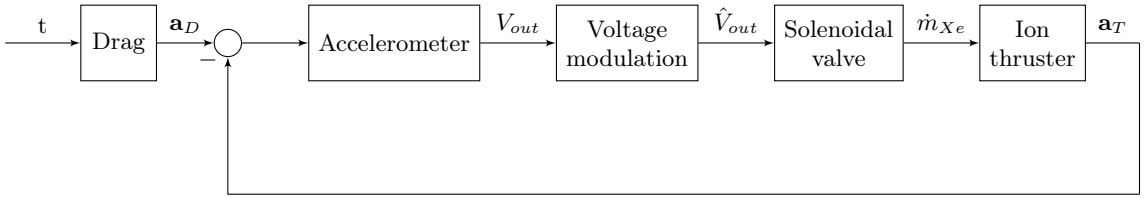


Figure 15: Block diagram of the ACS.

The ACS is modeled with several modules: an **accelerometer** measuring the accelerations acting on the spacecraft, a **voltage modulation block** controlling the output voltage, the **control valve** modifying the aperture of the thruster valve, and the **ion thruster**.

The accelerometer, kept aligned with the direction of velocity, presents inside a seismic mass that is subject to an external acceleration $\mathbf{a_D}$ and $\mathbf{a_T}$ given only by the drag D and the thrust T, respectively. The mass is in between the stators of a condenser and, moving for the acceleration given by these two forces changes the output voltage V_{out} of the circuit of the accelerometer. The dynamics of the seismic mass itself can be modeled as a mass-spring-damper system as follows

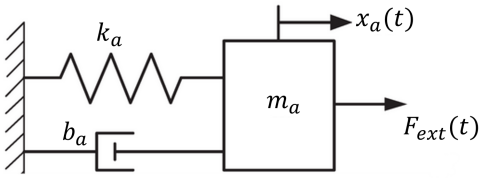


Figure 16: Model of the accelerometer.

$$\dot{x}_a = v_a \tag{11}$$

$$F_{ext}(t) = \frac{T - D}{M_{SC}} m_a = m_a \dot{v}_a + b_a v_a + k_a x_a \tag{12}$$

where M_{SC} is the mass of the spacecraft, m_a is the seismic mass value, and b_a and k_a are respectively the damper and the spring terms present in the accelerometer. For the causal modeling, consider the V_{out} as directly proportional to the velocity of the seismic mass, as $V_{out} = K_{acc} v_a$ [1].

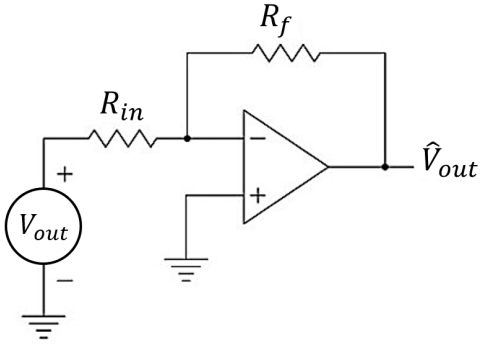


Figure 17: Inverting operational amplifier.

The output of the accelerometer V_{out} needs to be modulated into $\hat{V}_{out}$ to adjust the control for the solenoidal valve. This modulation is performed through an operational amplifier, displayed in Figure 17. The operational amplifier in inverting configuration modifies V_{out} into $\hat{V}_{out}$ accordingly to:

$$\hat{V}_{out} = -\frac{R_f}{R_{in}} V_{out} \tag{13}$$

where R_f and R_{in} are the two resistances modifying the input voltage.

The voltage $\hat{V}_{out}$ is itself the input of the solenoidal valve, shown in Figure 18, and creates a current I commanding the flow control valve. The current passes through a circuit modeled with only a solenoid with a variable inductance $L(x_v)$. As the current variably flows, a magnetic field is generated. A resulting force f_v acts on the spool, which as a result moves.

The armature-spool arrangement is then modeled again as a lumped parameter spring-mass-damper system

$$\dot{x}_v = v_v \tag{14}$$

$$m_v \dot{v}_v = -k_v x_v - b_v v_v + \frac{1}{2} I^2 \frac{dL}{dx_v} \tag{15}$$

$$\dot{I} = \frac{1}{L(x_v)} \hat{V}_{out} \tag{16}$$

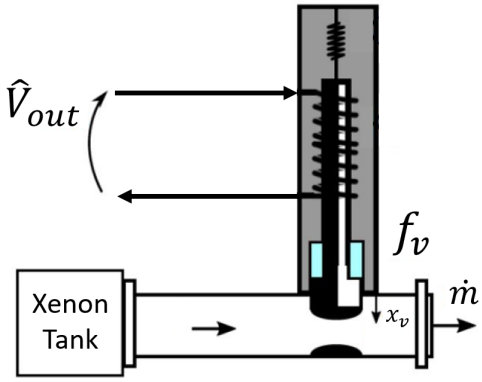


Figure 18: Model of the solenoidal valve.

where x_v and v_v represent the position and velocity of the spool, and b_v and k_v are respectively the damper and the spring terms, and m_v is the mass of the spool. The dynamics of the current I is generated by $\hat{V}_{out}$ through the variable inductance $L(x_v) = \frac{1}{\alpha + \beta x_v}$, depending on the position of the valve itself.

As the spool moves, the area $A_v(x_v)$ of the duct through which a Xenon flow is passing changes. The resulting mass flow rate $\dot{m}_{Xe}$ of Xenon is ejected into the ion thruster, shown in Figure 19. In particular, the area of the valve can be modeled as

$$A_v(x_v) = A_0 + \ell(x_{v,\max} - x_v) \tag{17}$$

In particular, the orifice area $A_v(x_v)$ has been modeled as linearly varying with x_v. The tool controlling the area is a flap of width ℓ and maximum extension $x_{v,\max} = \ell$, for simplicity.

The value A_0 is the minimum area of the orifice, such that:

$$0 \leq x_v \leq x_{v,\max}$$

$$\underbrace{A_0 = A_v(x_{v,\max})}_{\text{minimum area}} \leq A_v \leq \underbrace{A_v(0) = A_0 + \ell^2}_{\text{maximum area}} \tag{18}$$

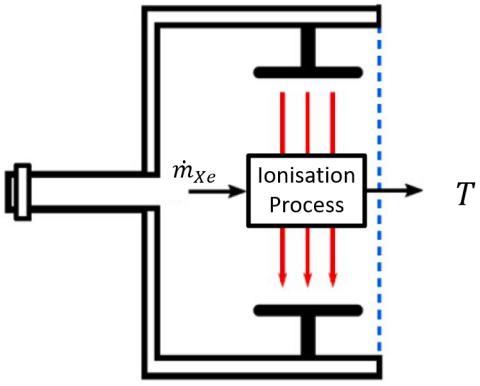


Figure 19: Model of the ion thruster.

The ion thruster comprises an ionization chamber that utilizes a strong magnetic field, ionizing the Xenon flux. The generated ions are moved towards the acceleration grid, which accelerates the ions using a strong electric field, producing a thrust on the spacecraft as the final effect. The flux of Xenon $\dot{m}_{Xe}$ can be expressed as follows

$$\dot{m}_{Xe} = A_v(x_v) \sqrt{k \rho_{Xe} p_T \left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}} \tag{19}$$

where $\rho_{Xe} = \frac{p_T}{\bar{R} T_T}$, k and $\bar{R}$ being respectively Xenon specific heat ratio and gas constant. In the tank, the gas is at a total pressure of p_T and temperature of T_T. The Xenon flow enters the

thruster and is here ionized into ions with mass m_i and charge q. After the acceleration imposed by the acceleration grid with a voltage ΔV, the ions exit from the nozzle with a velocity v_{exit}, producing the thrust T, as follows

$$T = \dot{m}_{Xe} v_{ext} = \dot{m}_{Xe} \sqrt{\frac{2q\Delta V}{m_i}} \tag{20}$$

Table 4 reports the values for the parameters to simulate the ACS system, with symbol and unit of measure. The drag $D(t)$ can be modeled as a function of time only, as it follows

$$D(t) = 2.2 - \cos(\omega_s t) + 1.2 \sin(\omega_o t) \cos(\omega_o t) \tag{21}$$

where t enters in seconds, and $D(t)$ is in mN.

Component	Parameter	Symbol	Value	Unit
Accelerometer	Spacecraft mass	M_{SC}	300	kg
	Seismic mass	m_a	0.32	kg
	Accelerometer damper	b_a	$[1.5{\cdot}10^3$–$2{\cdot}10^4]$	Ns/m
	Accelerometer spring	k_a	$[5{\cdot}10^{-5}$–$3{\cdot}10^{-3}]$	N/m
	Acc. proportional coefficient	K_{acc}	1	Vs/m
Amplifier	Inverting resistance	R_{in}	[0.1–10]	Ω
	Feedback resistance	R_f	$[1{\cdot}10^4$–$8{\cdot}10^4]$	Ω
Solenoidal Valve	Spool mass	m_v	0.1	kg
	Valve spring	k_v	$1{\cdot}10^3$	N/m
	Valve damper	b_v	$1{\cdot}10^3$	Ns/m
	Solenoid constant	α	$2.1{\cdot}10^{-2}$	1/H
	Solenoid gain	β	-60	1/Hm
	Minimum area	A_0	$4.7{\cdot}10^{-12}$	m^2
	Maximum extension	$x_{v,\max}$	$1{\cdot}10^{-5}$	m
Thruster	Heat ratio	k	1.66	-
	Tank pressure	p_T	$2{\cdot}10^5$	Pa
	Tank temperature	T_T	240	K
	Gas constant	$\bar{R}$	63.32754	J/kg K
	Charge	q	$1.6{\cdot}10^{-19}$	C
	Voltage	ΔV	2000	V
	Ion mass	m_i	$2.188{\cdot}10^{-25}$	kg
Drag	Secular pulsation	ω_s	$1.658226{\cdot}10^{-6}$	rad/s
	Orbital pulsation	ω_o	$1.160758{\cdot}10^{-3}$	rad/s

Table 4: Parameters for the simulation of the ACS.

Part 1: causal modeling (9 points)

Reproduce in Matlab the physical model of the ACS, in particular

1. Formulate the full system of non-linear ODEs making explicit the state variables of the whole simulation. Tune the values of the parameters not explicitly stated in Table 4 to have the compensating thrust T the most similar to the disturbing drag D.
2. Choose the appropriate initial conditions and simulate for three orbital periods T_o, where $T_o = 2\pi/\omega_o$.
3. Discuss the selection of the ODE integration scheme.

Part 2: acausal modeling (6 points)

Reproduce in Simscape the physical model of the ACS. Choose the appropriate initial conditions and parameters, simulate for three orbital periods T_o, and compare the results with those obtained in the prior point.

Note: when modeling the solenoid valve in Simscape, please consider the nominal values shown in Table 5.

Parameter	Value	Unit
Pull-in forces [F_1 - F_2]	[9000 - 12]	N
Stroke [x_1 - x_2]	[0 - 0.1]	mm
Maximum Stroke	0.1	mm
Rated voltage	0.6	mV
Rated current	0.1	A
Contact stiffness	0.12×10^6	N/m
Contact damping	10^4	Ns/m

Table 5: Solenoid parameters in Simscape.

(15 points)

2.1 Casual modeling

2.1.1 ODEs formulation & tuning

The system of non-linear ODEs is reported in Equation 23. The five state variables used are: position and velocity of the accelerometer (x_a, v_a), position and velocity of the spool (x_v, v_v) and the current flowing in the solenoid (I).
In order to write in a more compact formulation, the thrust (T) has been written as:

$$T = A(x_v)c_T \qquad \text{where} \qquad c_T = \sqrt{k\rho_{Xe}p_T(\frac{2}{k+1})^{\frac{k+1}{k-1}}}\sqrt{\frac{2q\Delta V}{m_i}} \tag{22}$$

This is because all the parameters inside c_T are constant.

$$\begin{cases} \dot{x_a} = v_a \\ \dot{v_a} = -\dfrac{k_a}{m_a}x_a - \dfrac{b_a}{m_a}v_a + \dfrac{(A_0 + \ell(x_{v,max} - x_v))c_T - D}{M_{SC}} \\ \dot{x_v} = v_v \\ \dot{v_v} = -\dfrac{k_v}{m_v}x_v - \dfrac{b_v}{m_v}v_v - \dfrac{1}{2}I^2\dfrac{\beta}{(\alpha + \beta x_v)m_v} \\ \dot{I} = -\dfrac{R_f}{R_i}K_{acc}v_a(\alpha + \beta x_v) \end{cases} \tag{23}$$

The tuned values of the parameters non explicit stated are reported in Table 6.

Parameter	Value	Unit
Accelerometer spring k_a	$5 \cdot 10^{-5}$	N/m
Accelerometer damper b_a	$1.5 \cdot 10^3$	Ns/m
Inverting resistance R_{in}	0.1	Ω
Feedback resistance R_f	$5 \cdot 10^4$	Ω

Table 6: Tuned parameters

2.1.2 Initial conditions

In order to simulate the system is necessary to select the appropriate set of initial conditions. Starting from the objective of the system, which is to compensate the drag force with the thrust, it is possible to find the value of the spool position at $t_0 = 0$ such that $T(t_0) = D(t_0)$. By imposing this condition and solving for x_v, the initial value of it is found. The velocity of the spool has been set to zero.
Then using the value found for the spool position is possible to determine the current value to have that position by substituting the previous values found of x_v and v_v inside Equation 15. For the position and velocity of the seismic mass (x_a, v_a) considering that the external force applied is equal to zero have been set equal to zero.

x_{a0} [m]	v_{a0} [m/s]	x_{v0} [m]	v_{v0} [m/s]	I_0 [A]
0	0	$8.584126e-6$	0	$-3.465153e-04$

Table 7: Initial conditions

2.1.3 Results

The system has been simulated for three orbital periods, where a single period is $T_o = 5314.002s \approx 1.50h$.
Considering the order of magnitude of all the variables strict tolerances have been used, the solver used is the stiff-solver `ode15s` because it allows to get an accurate solution with fewer steps and much lower computational time.
As can be observed by Figure 20 the ACS works successfully, in fact, the thrust is almost perfectly overlapped with the drag. Figure 21 shows the difference between the two forces it stays in the order of $1e-2\,mN$. with peaks in the order of $1e-5\,mN$, so can be concluded that the thrust managed to compensate the drag.

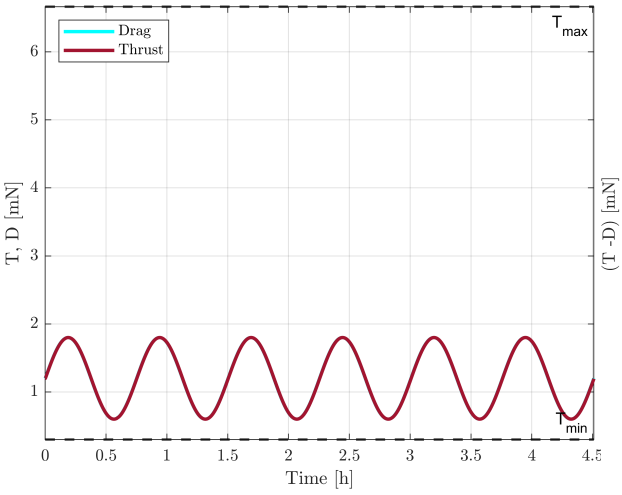


Figure 20: Time evolution of thrust and drag

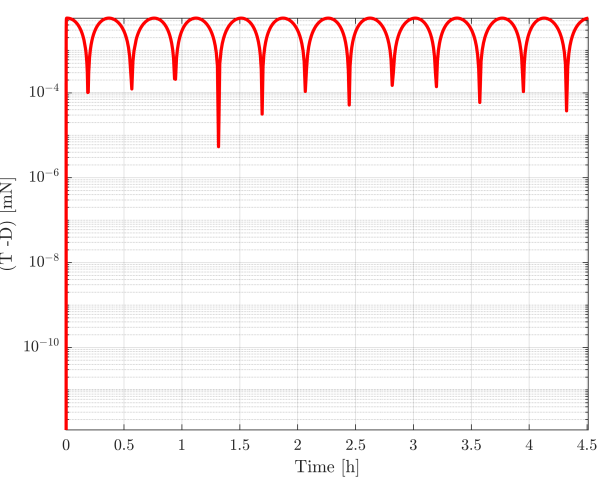


Figure 21: Evolution of $T - D$ over the time

By looking at the evolution of all the states can be observed that the global behavior is very similar, in fact for all of them the evolution is oscillatory. The main difference is in the velocity of the seismic mass and the spool,Figure 25 and Figure 23 respectively, is possible to observe that the behavior is very similar, both have a short transient and after an oscillatory evolution. Another consideration regards x_v, in fact as shown in Figure 24, the spool moves within the limits.

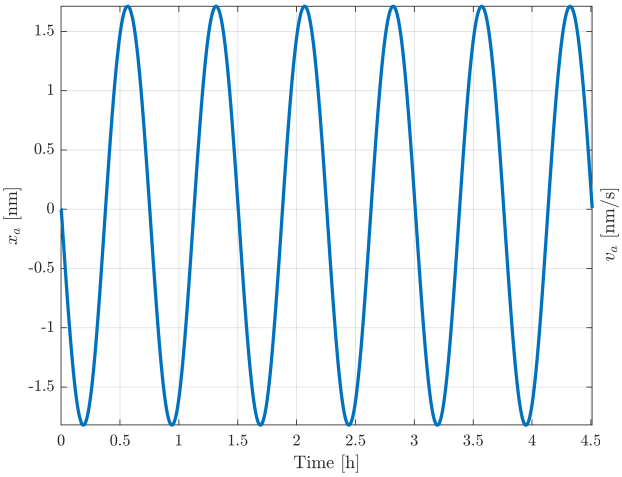


Figure 22: Seismic mass position evolution

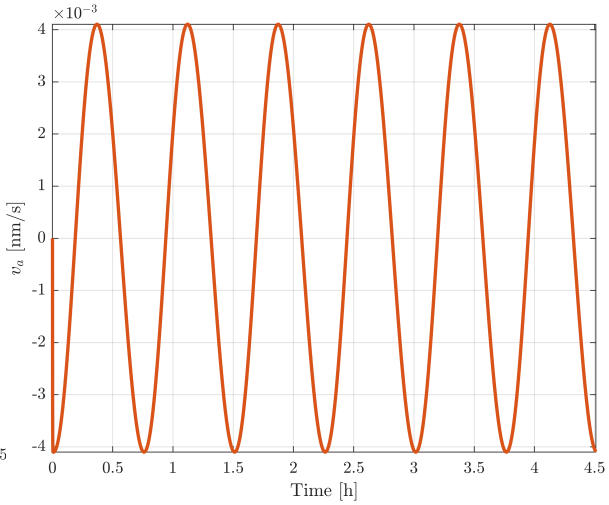


Figure 23: Seismic mass velocity evolution

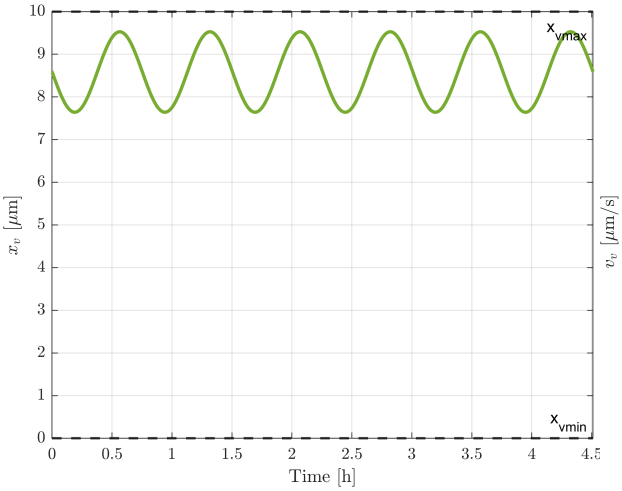


Figure 24: Spool position evolution

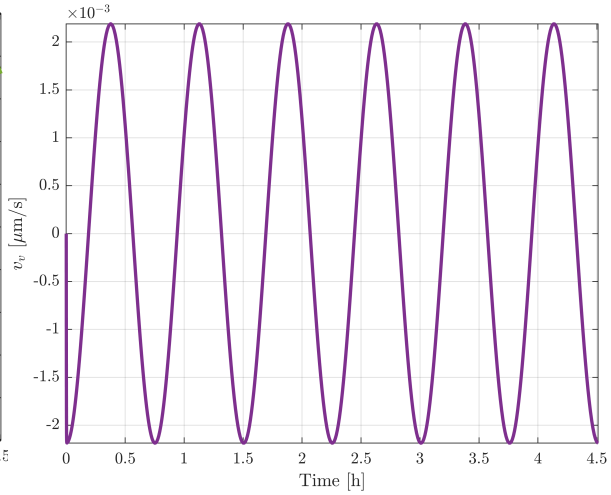


Figure 25: Spool velocity evolution

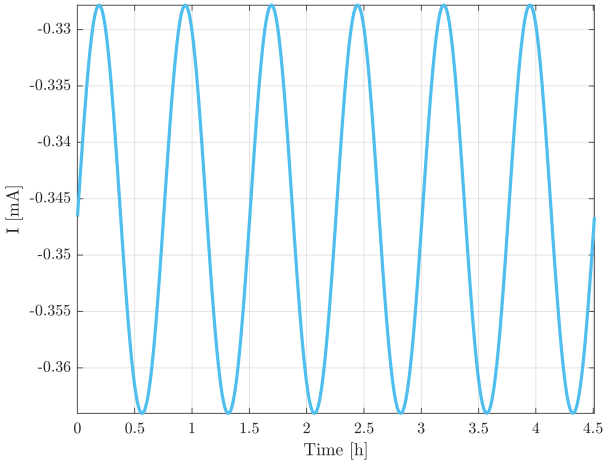


Figure 26: Current evolution

2.2 Acausal modeling

The model has been reproduced on Simscape, using the same parameter and initial conditions defined above.

The accelerometer has been modeled using the *Mass*, *Translational Spring*, and *Translational Damper* block the *Ideal Force Source* has been used to apply the external force on the mass. Then through the *Translational Electromechanical Converter* block the accelerometer interfaces with the voltage modulator. Using the *Solenoid* block the spool moves thanks to the modulated voltage.
After that using *Ideal Translational Motion Sensor* x_v is measured and goes as input in the block which computes $A(x_v)$ by multiplying it for the c_T through the *Gain* block the thrust is computed.
The blocks *To Workspace* are used to interface the Simscape model with MATLAB.

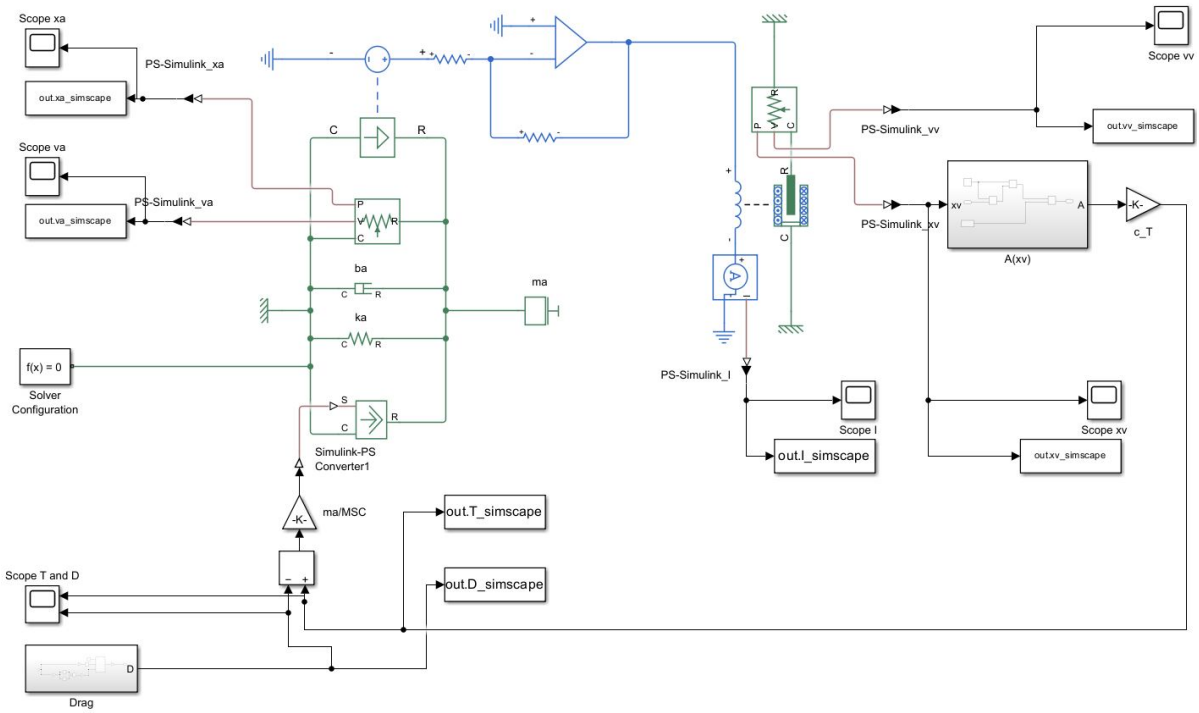


Figure 27: Simscape model

The Figure 28 and Figure 29 show that also in the solution obtained with Simscape the thrust balanced the drag.

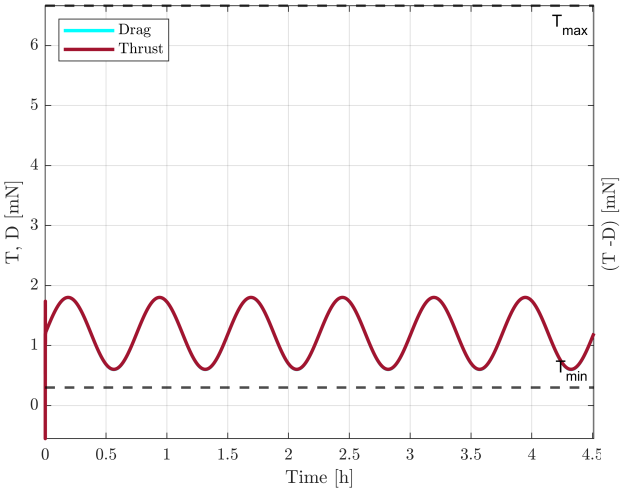


Figure 28: Time evolution of thrust and drag, obtained with Simscape

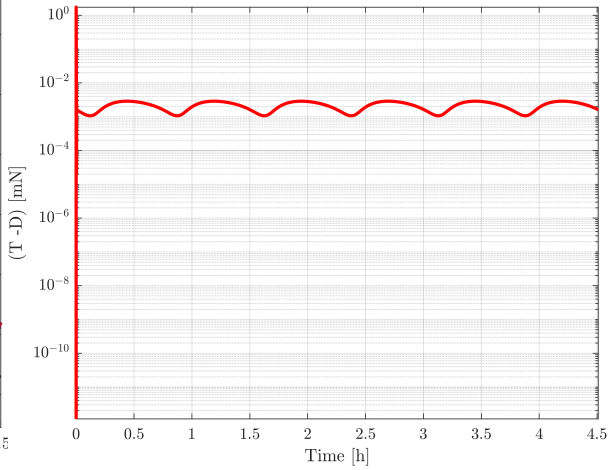


Figure 29: Evolution of $T - D$ over the time, obtained with Simscape

Even if in global terms the results are very similar, the two solutions are obtained using different solvers, so differences are present. The main difference is the transient of the states, in fact, all the states show a transient despite the initial condition. For x_v and v_v after the transient the behavior is the same, as shown in Figure 30 and Figure 31, as for MATLAB also in the Simscape solution x_v moves within the limits.

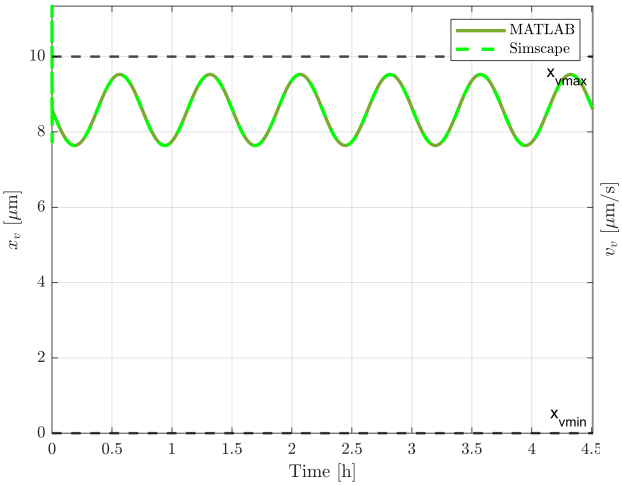


Figure 30: Comparison of the spool position evolution between MATLAB and Simscape

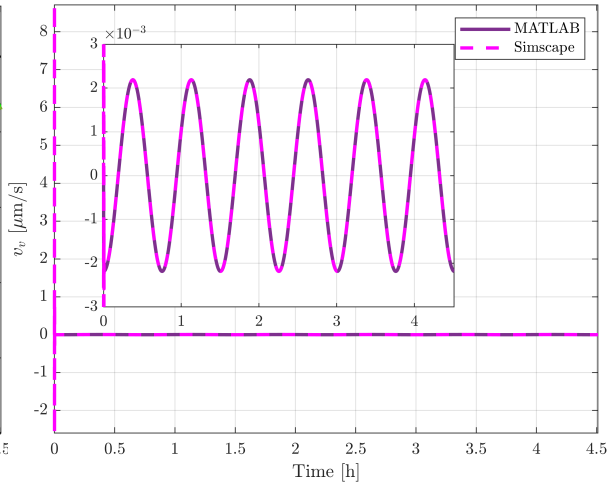


Figure 31: Comparison of the spool velocity evolution between MATLAB and Simscape

The main differences regards the x_a, v_a, I. The current oscillates with the same period and the peaks occur at the same time, but the nominal amplitude and the nominal value are different. The position of the seismic mass while does not oscillate around a nominal value but constantly grows, this is caused by v_a, due to the fact that the nominal value is different from zero, this behavior influences also the current, this happens because v_a appears in both ODEs.

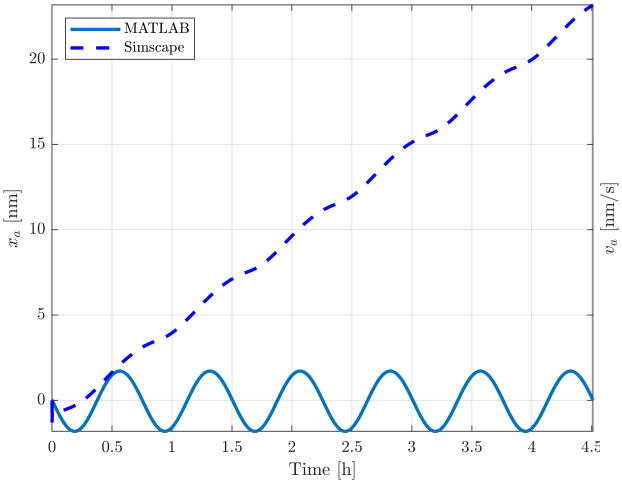


Figure 32: Comparison of the accelerometer's mass position evolution between MATLAB and Simscape

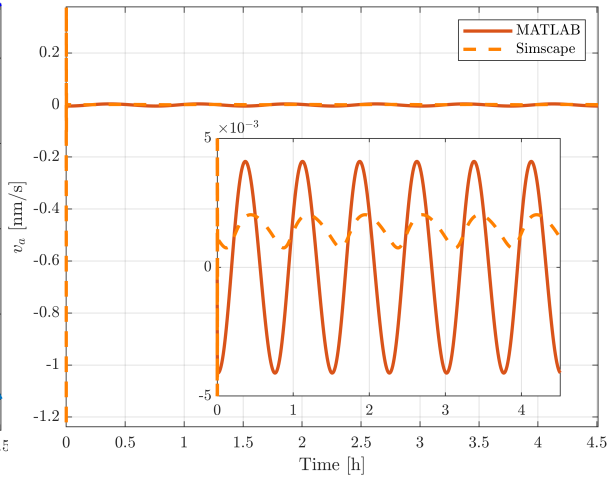


Figure 33: Comparison of the accelerometer's mass velocity evolution between MATLAB and Simscape

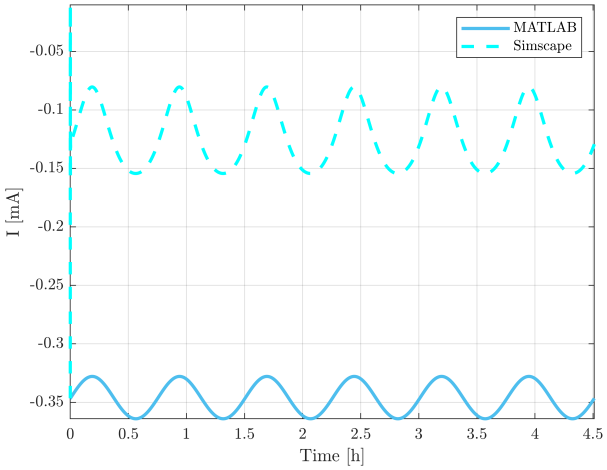


Figure 34: Comparison of the current evolution between MATLAB and Simscape

The last considerations can be made about the T. By looking at the difference between T and D in logarithmic scale, Figure 35, can be observed that for both methods the difference is in the order of $1e-2\,mN$. Instead considering the Figure 36 can be noticed that the mean value of $T-D$ computed with MATLAB is zero, while the Simscape one is a positive value, so T is always greater than the drag. This shift in the mean value arises primarily due to the behavior of v_a.

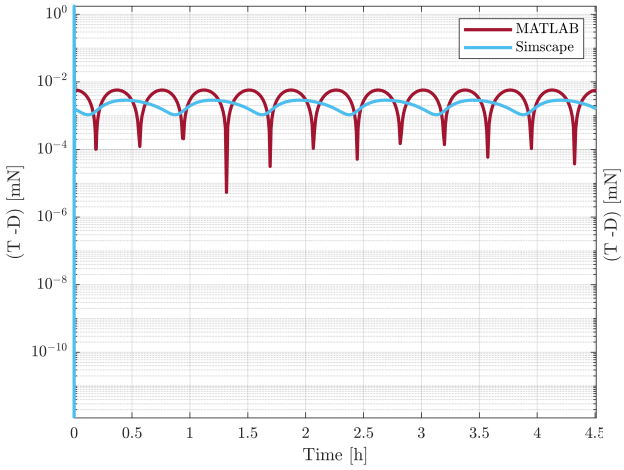


Figure 35: Comparison between the difference between thrust and drag for MATLAB and Simscape in logarithmic scale

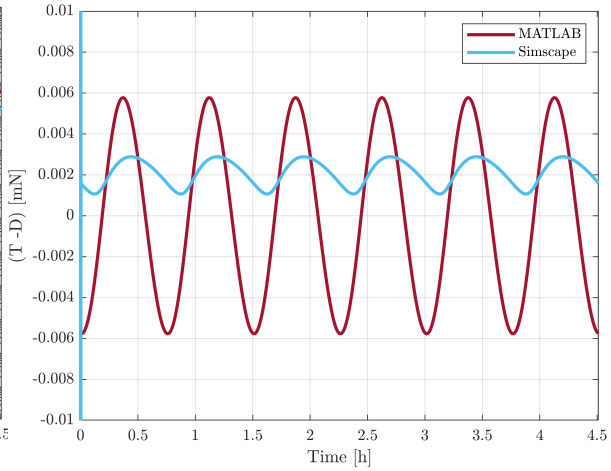


Figure 36: Comparison between the difference between thrust and drag for MATLAB and Simscape

References

[1] R. Mukhiya, M. Garg, P. Gaikwad, et al. *Electrical equivalent modeling of MEMS differential capacitive accelerometer*. In: Microelectronics Journal. Vol. 99, pp. 104770, 2020. `https://doi.org/10.1016/j.mejo.2020.104770`